

Experimental VLE, consistency and fitting

After midterm · Session 2 of 6 · 180 minutes

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The question for today

Does a small fitting error mean the data and model are trustworthy?

Before class: 3-page reading · 15–20 minutes

Bring the three preparation answers. Revisit your prediction after one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion, exit ticket and feedback	30

The 70-minute investigation is student work. Final block: discussion 20 min, exit ticket 5 min, feedback 3 min, closing 2 min.

Activity and its reference

$$\mu_i = \mu_i^\circ + RT \ln a_i, \quad a_i = \gamma_i x_i$$

The activity is dimensionless; γ is its correction relative to an ideal liquid solution using the pure-liquid standard.

Residual: real fluid minus ideal gas at the same T, P and composition.

Excess: real mixture minus ideal solution at the same T, P and composition.

Properties → prediction → fitting

1. Verify the Antoine form, units and applicable temperature range.
2. Predict P and y from a chosen liquid model.
3. Infer activities from observations, then compare model fits.

$$\gamma_i = \frac{y_i P}{x_i P_i^{sat}(T)}$$

This inversion assumes ideal vapor and negligible Poynting correction. Convergence, consistency and prediction are separate checks.

A vapor-pressure error becomes an activity error

Synthetic point: $P=100$ kPa, $x_1=0.4$, $y_1=0.6$ and $P_1^{\text{sat}}=120$ kPa.

$$\gamma_1 = \frac{0.6(100)}{0.4(120)} = 1.25$$

Using 150 kPa instead yields $\gamma_1=1.00$. The measured mixture data did not change; the inferred liquid behavior did.

An independent check

Check one Antoine saturation pressure and one inferred activity coefficient by hand. Keep the same observations and property convention when comparing models.

One demonstration • 20 minutes

[Open the core lab](#)

Load and validate the synthetic example in Lab 08. Inspect its Antoine form and units, then fit one-parameter Margules. Compare predicted P and y with one observed point, then inspect the residual pattern.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 08, validate the supplied synthetic example and compare one-parameter Margules with NRTL at $\alpha=0.3$ on the same observations. Report one residual pattern and verify one activity coefficient by hand.

Discussion • 20 minutes

Does a small fitting error mean the data and model are trustworthy?

Compare results and discuss unexpected behavior. Use one numerical check to support your explanation.

After class · 30–40 minutes

In Lab 08, validate the supplied synthetic example and compare one-parameter Margules with NRTL at $\alpha=0.3$ on the same observations. Report one residual pattern and verify one activity coefficient by hand.

Assessment 2 and independent checkpoint

Retain one study JSON, one comparison and a short explanation.

Optional: Session appendix.

Learning path · Offline package

Exit ticket + feedback · 8 minutes

5 min · Explain and check

Answer the session question in the form. Give one result or independent check from today's lab, with units.

3 min · Help improve the next class

Rate the pace, report lab access, and suggest one improvement.

Thai or English. No name, ID or email requested.

Wait for **“Response received.”** We will use the final 2 minutes to close the class.



Scan to submit Session 2 exit ticket and feedback.

[Open Session 2 form](#)